



SILICON UNIVERSITY

SKILL LAB & PROJECT -1

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PROJECT: EcoStreet (Smart Street Light)

EEE-B1, GROUP-01

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We are equally thankful to **Mr. Sanjaya Mahapatra** for his practical mentorship, constant motivation, and real-world insights. His suggestions helped us improve the system's scalability, sensor integration, and overall reliability—ensuring that EcoStreet is both sustainable and adaptable to future smart city needs.

Their combined mentorship was instrumental in shaping our understanding of how smart technology can address real-world urban challenges, reduce energy consumption, and contribute toward building smarter, greener communities.

ABSTRACT

The *EcoStreet – Smart Street Light* project aims to develop an intelligent and energy-efficient lighting system that responds dynamically to environmental conditions and human activity. Using sensors such as LDR (Light Dependent Resistor) and IR (Infrared), the system automatically adjusts brightness based on ambient light and motion detection. Built on the Arduino Uno platform, EcoStreet minimizes energy consumption, reduces operational costs, and supports sustainable urban development. This project demonstrates how embedded systems and IoT technologies can enhance public safety while promoting eco-friendly infrastructure solutions.

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INTRODUCTION

With the rapid growth of urban areas, traditional street lighting systems have become inefficient, consuming excessive electricity and requiring high maintenance. The *EcoStreet – Smart Street Light* project introduces a smarter solution by using sensors and microcontrollers to automate lighting based on real-time environmental conditions and human presence. This project supports sustainable development by reducing power consumption, minimizing manual intervention, and enhancing public safety.

EQUIPMENT REQUIRED

The *EcoStreet – Smart Street Light* system is built using a combination of simple yet effective electronic components that work together to enable smart, energy-efficient lighting. The core of the system is the **Arduino Uno**, a popular microcontroller board that serves as the brain of the project. It processes sensor inputs and controls the lighting output accordingly.

To detect the surrounding light intensity, an **LDR (Light Dependent Resistor)** is used. It helps determine whether it is day or night and adjusts the streetlight status accordingly. For motion detection, **IR (Infrared) sensors** are employed. These sensors detect the presence of vehicles or pedestrians and trigger the light to brighten when motion is detected and return to a dim state when the area is clear.

LEDs or a streetlight module act as the actual light source in the prototype, representing real-world streetlamps. Additional components such as **resistors** ensure safe

current flow, while a **breadboard and jumper wires** provide a temporary platform for connecting and testing the circuit components before final implementation.

Together, these components form a cost-effective, responsive, and eco-friendly lighting system that can be easily integrated into urban infrastructure.

Discussing the equipment used in detail:

1. Arduino Uno

The Arduino Uno is a versatile microcontroller board based on the ATmega328P chip, and it forms the central control unit of the smart streetlight system. It is programmed to receive signals from sensors and control the LED lighting accordingly. With multiple digital and analog I/O pins, it enables easy interfacing with components like IR sensors and LDRs. Its open-source platform and USB connectivity make it ideal for rapid prototyping and student projects. In EcoStreet, it ensures automated decisions for lighting based on input conditions.

2. LDR (Light Dependent Resistor)

An LDR is a photoresistor that detects the intensity of light in the environment. Its resistance decreases as the light level increases, allowing it to act as a sensor for distinguishing between day and night. In the smart streetlight system, the LDR informs the Arduino when it becomes dark, triggering the system to activate lighting. It plays a crucial role in energy conservation by ensuring that the streetlights remain off during the day and operate only at night or in low-light conditions.

3. IR Sensor

Infrared (IR) sensors are used to detect motion in the vicinity of the streetlight. These sensors emit infrared light and detect the reflection from nearby objects, allowing them to sense movement of vehicles or pedestrians. In this project, IR sensors are placed at intervals to provide real-time detection. When motion is detected, the sensor sends a signal to the Arduino, which then increases the

brightness of the LED streetlight. Once the motion stops, the lights return to a dim state, optimizing energy usage.

4. LEDs (Streetlight Simulation)

LEDs are used in this project to simulate real streetlights. They are energy-efficient, bright, and long-lasting, making them ideal for modern lighting solutions. The brightness of the LEDs can be adjusted using pulse-width modulation (PWM) from the Arduino. In the smart streetlight system, LEDs turn on in a dim state at night and brighten only when motion is detected nearby. This approach saves power while maintaining adequate illumination for safety.

5. Resistors

Resistors are essential for regulating current in the circuit and protecting sensitive components. In the smart streetlight setup, they are often used with LEDs and sensors to ensure that they operate within safe voltage and current limits. For example, in the LDR circuit, resistors help form a voltage divider to provide accurate analog readings to the Arduino. Their role, though passive,

is critical in maintaining circuit stability and preventing damage to components.

6. Breadboard and Jumper Wires

The breadboard is used for building and testing the circuit without soldering. It allows all components to be connected easily in a temporary layout for prototyping. Jumper wires are used to make these electrical connections between the Arduino, sensors, LEDs, and other components. This combination helps in rapid testing and modification during the development phase. Once the design is finalized, it can be moved to a permanent PCB or enclosure for deployment.

WORKING

The smart street light system operates using sensors and a microcontroller (Arduino Uno) to automate lighting based on environmental and motion inputs:

1. **Daylight Detection (LDR):**

The LDR measures ambient light. During daytime, the light intensity is high, so the Arduino keeps the LEDs off to save energy.

2. **Night-Time Activation:**

When it gets dark (low light detected by LDR), the Arduino turns the LEDs on in a **dim state**.

3. **Motion Detection (IR Sensor):**

When a vehicle or pedestrian passes by, the IR sensor detects motion and sends a signal to the Arduino.

4. **LED Brightening:**

Upon detecting motion, the Arduino increases

the LED brightness to **full intensity**, providing better visibility and safety.

5. **Auto Dimming:**

After a few seconds of no movement, the lights automatically return to dim mode to conserve power.

This cycle repeats, making the system fully **automated, energy-efficient, and suitable for smart city infrastructure.**

CONCLUSION

The *EcoStreet – Smart Street Light* project successfully demonstrates how simple electronic components and smart automation can solve real-world urban challenges. By combining motion detection and ambient light sensing, the system optimizes energy usage while enhancing safety and reducing operational costs. This intelligent approach not only supports environmental sustainability but also paves the way for future-ready smart city infrastructure. The project reflects innovation, practicality, and a clear step toward a greener, more efficient future — proving that small systems can lead to impactful change in modern society.

*THANK
YOU*